

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTECPE405A - MICROCONTROLLER AND ADVANCED PROCESSOR

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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1. What is the size of the Program Counter (PC) in 8051?
a) 8-bit b) 10-bit c) 12-bit d) 16-bit
2. In 8051, mode 2 of Timer works as:
a) 13-bit timer b) 16-bit timer c) Auto-reload d) Counter only
3. How many interrupt sources are available in 8051?
a) 3 b) 5 c) 6 d) 8
4. Which interrupt is used for serial data reception in 8051?
a) INT0 b) INT1 c) RI d) TI
5. 8051 microcontroller has how many 16 bit registers?
a) 2 b) 3 c) 1 d) 0
6. In LCD interfacing, which pin is used to select between command and data register?
a) RS b) RW c) EN d) VSS
7. The ADC0808 has how many analog input channels?
a) 4 b) 6 c) 8 d) 10
8. Which of the following is not a mode of operation in ARM7?
a) Supervisor b) Abort c) Monitor d) Undefined
9. In ARM 7 architecture, which register stores the return address after a subroutine call?
a) SP b) R15 c) LR d) PC
10. The LPC2148 microcontroller is based on which processor core?
a) ARM Cortex-M4 b) ARM7TDMI-S c) ARM9E d) Cortex-M3
11. LPC 2148 pro development board has _____ on chip memory.
a) 500k b) 625k c) 512k d) 425k
12. LPC2148 supports how many 32-bit timers?
a) 1 b) 2 c) 4 d) 0

Q2. Solve the following.

- A) Explain the differences between Harvard and Von Neumann architectures.**

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- B) Describe memory organization of the 8051 microcontroller. 6
- Q3. Solve the following.
- A) Write an 8051 assembly language program to generate a square wave of frequency 6 10 KHz on pin P1.0. Assume that XTAL = 11.0592 MHz.
- B) Draw an interfacing diagram to interface a 7 segment display to 8051 6 microcontroller and write an ALP to display '0'.
- Q4. Solve Any Two of the following.
- A) Draw an interfacing diagram to interface a relay to 8051 microcontroller and 6 write an ALP to turn it ON and OFF with certain time delay.
- B) Write an 8051 assembly program to rotate a stepper motor clockwise using Port 2. 6
- C) Describe the process of interfacing DAC0808 with 8051 and write an ALP to 6 display a triangular waveform.
- Q5. Solve Any Two of the following.
- A) Explain significance of each bit of flag register of ARM 7 processor. 6
- B) Discuss all the modes of operation of ARM 7 processor. 6
- C) Describe ARM7 processor data flow model. 6
- Q6. Solve Any Two of the following.
- A) Explain memory map of LPC2148 microcontroller. 6
- B) Draw an architecture of LPC2148 microcontroller and list out its features. 6
- C) Write a note on renesas microcontroller RL78/G13. 6

*** End ***